

2 The modular character of elliptic functions and Eisenstein series

We shall now study the modular character of elliptic functions, that is, their dependence on τ .

Recall the normalization we made at the beginning of the chapter. We started with two periods ω_1 and ω_2 linearly that are independent over $\mathbb{R}$, and we defined $\tau = \omega_2/\omega_1$. We could then assume that $\text{Im}(\tau) > 0$, and also that the two periods are 1 and τ . Next, we considered the lattice generated by 1 and τ and constructed the function $\wp$, which is elliptic of order 2 with periods 1 and τ . Since the construction of $\wp$ depends on τ , we could write $\wp_\tau$ instead. This leads us to change our point of view and think of $\wp_\tau(z)$ primarily as a function of τ . This approach yields many interesting new insights.

Our considerations are guided by the following observations. First, since 1 and τ generate the periods of $\wp_\tau(z)$, and 1 and $\tau + 1$ generate the same periods, we can expect a close relationship between $\wp_\tau(z)$ and $\wp_{\tau+1}(z)$. In fact, it is easy to see that they are identical. Second, since $\tau = \omega_2/\omega_1$, by the normalization imposed at the beginning of Section 1, we see that $-1/\tau = -\omega_1/\omega_2$ (with $\text{Im}(-1/\tau) > 0$). This corresponds essentially to an interchange of the two periods ω_1 and ω_2 , and thus we can also expect an intimate connection between $\wp_\tau$ and $\wp_{-1/\tau}$. In fact, it is easy to verify that $\wp_{-1/\tau}(z) = \tau^2 \wp_\tau(\tau z)$.

So we are led to consider the group of transformations of the upper half-plane $\text{Im}(\tau) > 0$, generated by the two transformations $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$. This group is called the **modular group**. On the basis of what we said, it can be expected that all quantities intrinsically attached to $\wp_\tau(z)$ reflect the above transformations. We see this clearly when we consider the Eisenstein series.

2.1 Eisenstein series

The **Eisenstein series** of order k is defined by

$$E_k(\tau) = \sum_{(n,m) \neq (0,0)} \frac{1}{(n + m\tau)^k},$$

whenever k is an integer ≥ 3 and τ is a complex number with $\text{Im}(\tau) > 0$. If Λ is the lattice generated by 1 and τ , and if we write $\omega = n + m\tau$, then another expression for the Eisenstein series is $\sum_{\omega \in \Lambda^*} 1/\omega^k$.

Theorem 2.1 *Eisenstein series have the following properties:*

- (i) The series $E_k(\tau)$ converges if $k \geq 3$, and is holomorphic in the upper half-plane.
- (ii) $E_k(\tau) = 0$ if k is odd.
- (iii) $E_k(\tau)$ satisfies the following transformation relations:

$$E_k(\tau + 1) = E_k(\tau) \quad \text{and} \quad E_k(\tau) = \tau^{-k} E_k(-1/\tau).$$

The last property is sometimes referred to as the **modular** character of the Eisenstein series. We shall return to these and other modular identities in the next chapter.

Proof. By Lemma 1.5 and the remark after it, the series $E_k(\tau)$ converges absolutely and uniformly in every half-plane $\text{Im}(\tau) \geq \delta > 0$, whenever $k \geq 3$; hence $E_k(\tau)$ is holomorphic in the upper half-plane $\text{Im}(\tau) > 0$.

By symmetry, replacing n and m by $-n$ and $-m$, we see that whenever k is odd the Eisenstein series is identically zero.

Finally, the fact that $E_k(\tau)$ is periodic of period 1 is clear from the fact that $n + m(\tau + 1) = n + m + m\tau$, and that we can rearrange the sum by replacing $n + m$ by n . Also, we have

$$(n + m(-1/\tau))^k = \tau^{-k} (n\tau - m)^k,$$

and again we can rearrange the sum, this time replacing $(-m, n)$ by (n, m) . Conclusion (iii) then follows.

Remark. Because of the second property, some authors define the Eisenstein series of order k to be $\sum_{(n,m) \neq (0,0)} 1/(n + m\tau)^{2k}$, possibly also with a constant factor in front.

The connection of the E_k with the Weierstrass $\wp$ function arises when we investigate the series expansion of $\wp$ near 0.

Theorem 2.2 For z near 0, we have

$$\begin{aligned} \wp(z) &= \frac{1}{z^2} + 3E_4 z^2 + 5E_6 z^4 + \cdots \\ &= \frac{1}{z^2} + \sum_{k=1}^{\infty} (2k+1) E_{2k+2} z^{2k}. \end{aligned}$$

Proof. From the definition of $\wp$, if we note that we may replace ω by $-\omega$ without changing the sum, we have

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right] = \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right],$$

where $\omega = n + m\tau$. The identity

$$\frac{1}{(1-w)^2} = \sum_{\ell=0}^{\infty} (\ell+1)w^{\ell}, \quad \text{for } |w| < 1,$$

which follows from differentiating the geometric series, implies that for all small z

$$\frac{1}{(z-\omega)^2} = \frac{1}{\omega^2} \sum_{\ell=0}^{\infty} (\ell+1) \left(\frac{z}{\omega}\right)^{\ell} = \frac{1}{\omega^2} + \frac{1}{\omega^2} \sum_{\ell=1}^{\infty} (\ell+1) \left(\frac{z}{\omega}\right)^{\ell}.$$

Therefore

$$\begin{aligned} \wp(z) &= \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \sum_{\ell=1}^{\infty} (\ell+1) \frac{z^{\ell}}{\omega^{\ell+2}} \\ &= \frac{1}{z^2} + \sum_{\ell=1}^{\infty} (\ell+1) \left(\sum_{\omega \in \Lambda^*} \frac{1}{\omega^{\ell+2}} \right) z^{\ell} \\ &= \frac{1}{z^2} + \sum_{\ell=1}^{\infty} (\ell+1) E_{\ell+2} z^{\ell} \\ &= \frac{1}{z^2} + \sum_{k=1}^{\infty} (2k+1) E_{2k+2} z^{2k}, \end{aligned}$$

where we have used the fact that $E_{\ell+2} = 0$ whenever ℓ is odd.

From this theorem, we obtain the following three expansions for z near 0:

$$\begin{aligned} \wp'(z) &= \frac{-2}{z^3} + 6E_4 z + 20E_6 z^3 + \cdots, \\ (\wp'(z))^2 &= \frac{4}{z^6} - \frac{24E_4}{z^2} - 80E_6 + \cdots, \\ (\wp(z))^3 &= \frac{1}{z^6} + \frac{9E_4}{z^2} + 15E_6 + \cdots. \end{aligned}$$

From these, one sees that the difference $(\wp'(z))^2 - 4(\wp(z))^3 + 60E_4\wp(z) + 140E_6$ is holomorphic near 0, and in fact equal to 0 at the origin. Since this difference is also doubly periodic, we conclude by Theorem 1.2 that it is constant, and hence identically 0. This proves the following corollary.

Corollary 2.3 *If $g_2 = 60E_4$ and $g_3 = 140E_6$, then*

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3.$$

Note that this identity is another version of Theorem 1.7, and it allows one to express the symmetric functions of the e_j 's in terms of the Eisenstein series.

2.2 Eisenstein series and divisor functions

We will describe now the link between Eisenstein series and some number-theoretic quantities. This relation comes about if we consider the Fourier coefficients in the Fourier expansion of the periodic function $E_k(\tau)$. Equivalently, we can write $\mathcal{E}(z) = E_k(\tau)$ with $z = e^{2\pi i\tau}$, and investigate the Laurent expansion of $\mathcal{E}$ as a function of z .

We begin with a lemma.

Lemma 2.4 *If $k \geq 2$ and $\text{Im}(\tau) > 0$, then*

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n + \tau)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i\tau\ell}.$$

Proof. This identity follows from applying the Poisson summation formula to $f(z) = 1/(z + \tau)^k$; see Exercise 7 in Chapter 4.

An alternate proof consists of noting that it first suffices to establish the formula for $k = 2$, since the other cases are then obtained by differentiating term by term. To prove this special case, we differentiate the formula for the cotangent derived in Chapter 5

$$\sum_{n=-\infty}^{\infty} \frac{1}{n + \tau} = \pi \cot \pi\tau.$$

This yields

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n + \tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

Now use Euler's formula for the sine and the fact that

$$\sum_{r=1}^{\infty} r w^r = \frac{w}{(1-w)^2} \quad \text{with } w = e^{2\pi i\tau}$$

to obtain the desired result.

As a consequence of this lemma, we can draw a connection between the Eisenstein series, the zeta function, and the divisor functions. The